BMS_SLAVE

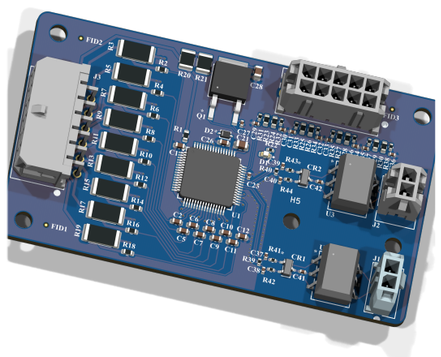
Variant: PRELIMINARY

2026-10-12

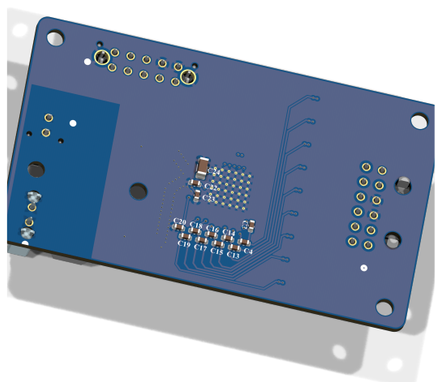
Rev + (Unreleased)

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11	.....	21	.....	31	.....
2	Block Diagram	12	.....	22	.....	32	.....
3	Project Architecture	13	.....	23	.....	33	.....
4	Connectors	14	.....	24	.....	34	.....
5	Cell Sense & Balance	15	.....	25	.....	35	.....
6	Power & Reference	16	.....	26	.....	36	.....
7	Temperature Sense	17	.....	27	.....	37	.....
8	Daisy Chain	18	.....	28	.....	38	.....
9	Mechanical	19	.....	29	.....	39	.....
10	Revision History	20	.....	30	.....	40	.....

TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELESED - A board with this schematic has been sent to production.

Date: 12-Oct-2026

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

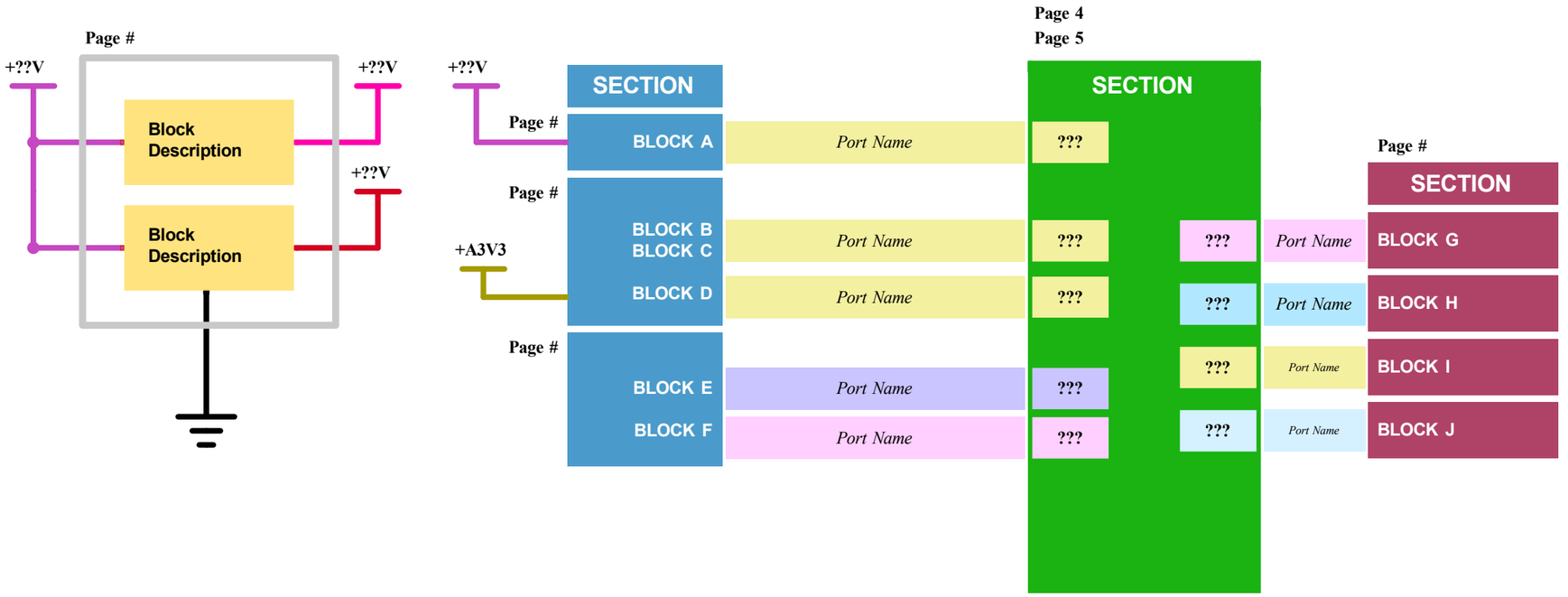
DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

Comments:	Company: Blue Devil Motor Sports			Variant:
	Board Name: BMS_SLAVE			Project Name: BMS Slave
Sheet Title: bms_slave	File Name: bms_slave.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision: + (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 10

[2] Block Diagram

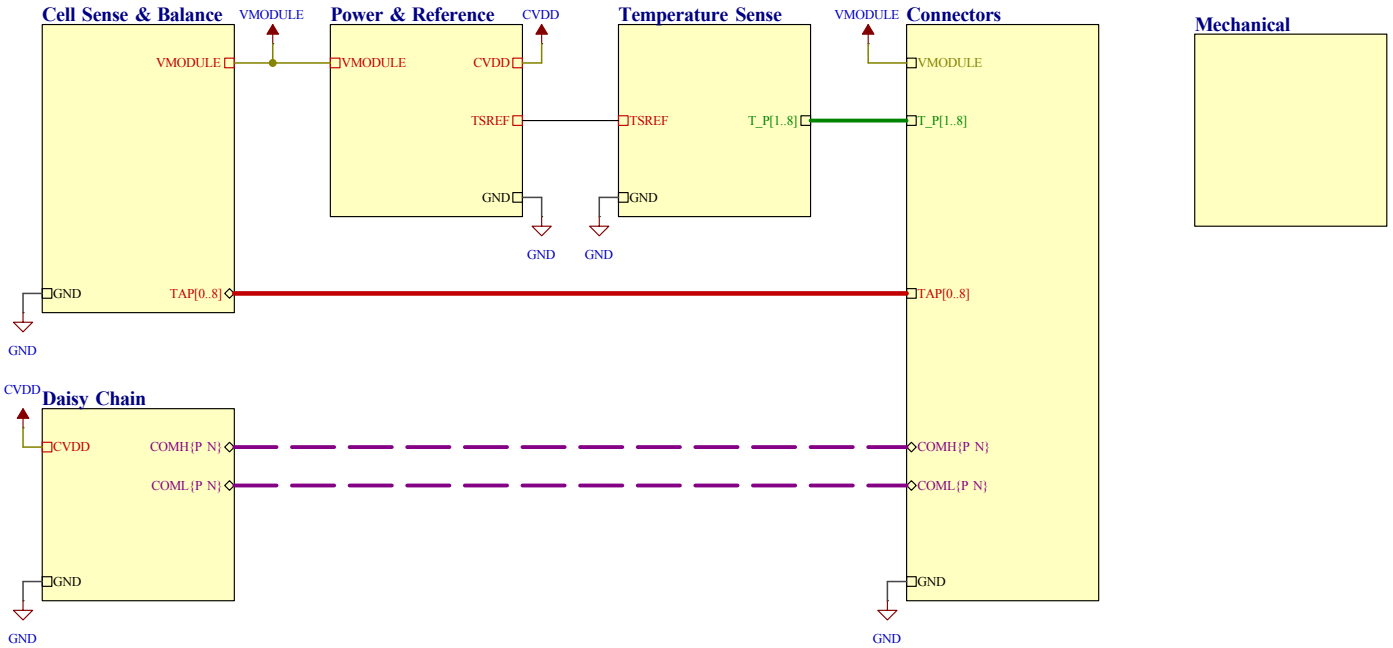


Target specifications:

Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

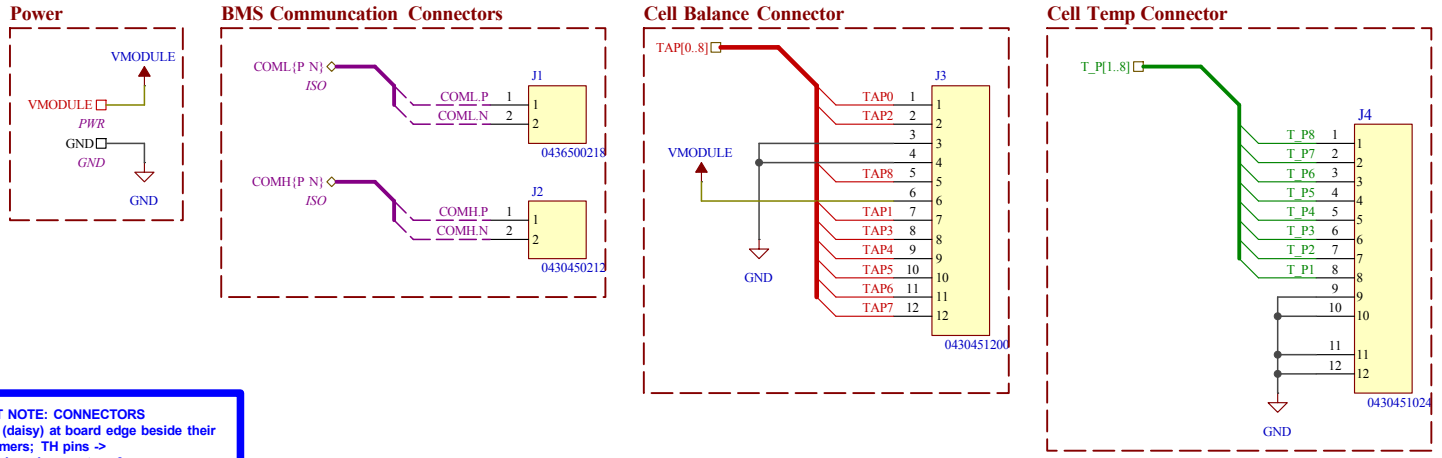
Comments:	Company: **Blue Devil Motor Sports**		Variant:	
	Board Name: **BMS_SLAVE**		Project Name: **BMS Slave**	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision: + (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: **A3**	Sheet: **2** of **10**

[3] Project Architecture



Comments:	Company: **Blue Devil Motor Sports**		Variant:	
	Board Name: **BMS_SLAVE**		Project Name: **BMS Slave**	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision: + (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: **A4**	Sheet: **3** of **10**

[4] Connectors



DESIGN NOTE:
Using a single ground on connector assumes that on the battery interface board there is going to be a common ground for all temp wires.

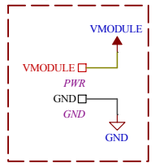
LAYOUT NOTE: CONNECTORS
1. J1/J4 (daisy) at board edge beside their transformers; TH pins -> inner-plane keep-out >= 3 mm.
2. J2 taps / J3 temps on the edge facing the blocks.
3. Tab fuses at CELL END of tap harness, >= 67.2 V rated (FSAE EV.8.4.3).
4. Verify pin 1 / row numbering on each Molex drawing.

Comments:	Company:		Variant:	
	Board Name: **BMS_SLAVE**		Project Name: **BMS Slave**	
Sheet Title: Connectors	File Name: connectors.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
Sheet Path: /Project Architecture/Connectors/		Reviewer:	Size: **A4**	Sheet: **4** of **10**

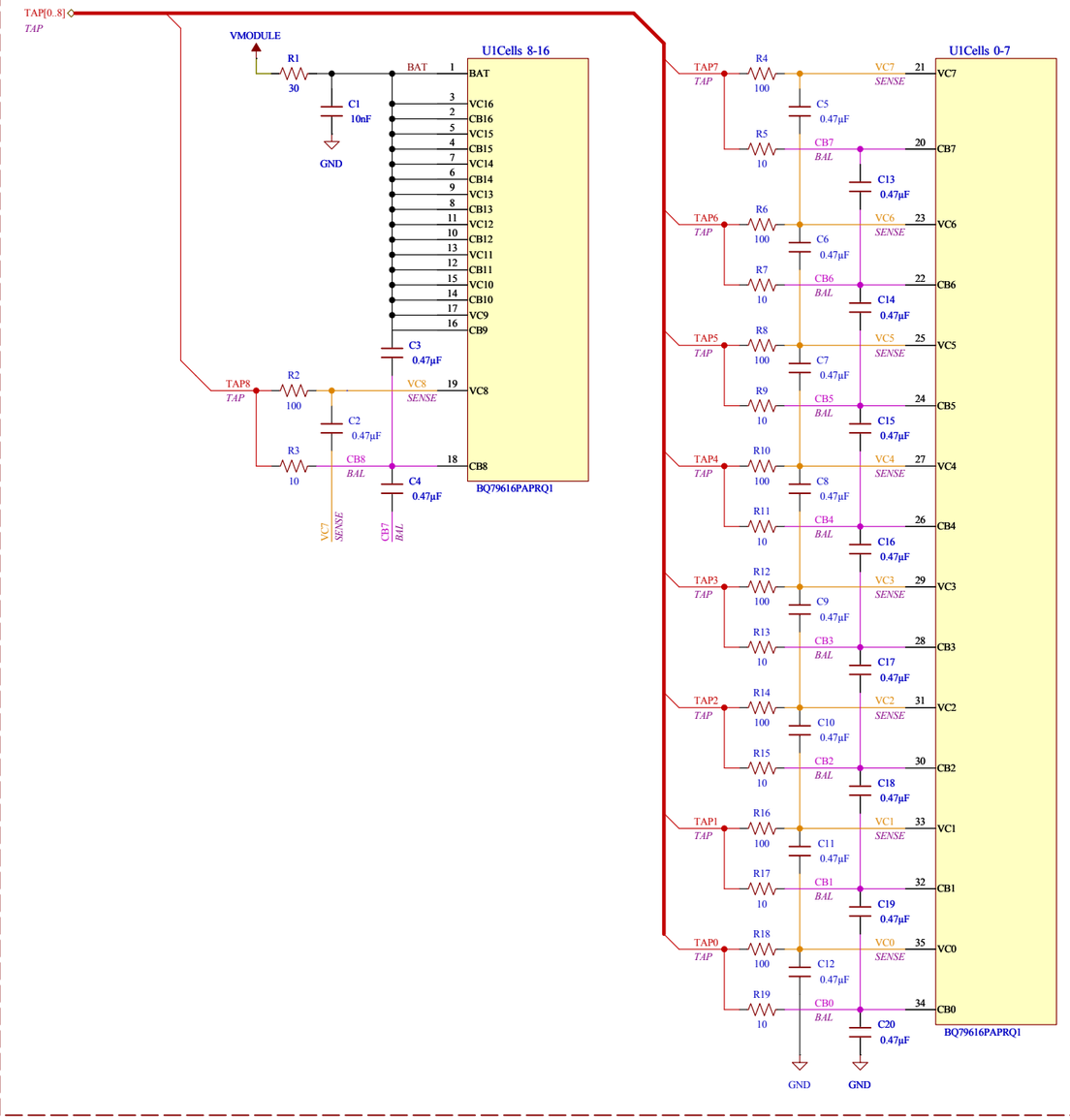
[5] Cell Sense & Balance

LAYOUT NOTE: CELL SENSE (SLUSE81F)
1. VC/CB filter caps at U1 pins; RC values per 8.2.1.2.1 (p.192).
2. VC and CB traces run in parallel, impedance matched; CB traces sized for ~190 mA balancing (10.1.3, p.205).
3. Unused VC9-16/CB9-16 tie to BAT pin with minimum trace; never to a wire or cable (Fig. 8-2a, p.192).
4. Separate the BAT/VMODULE branch from TAP8, and the GND branch from TAP0, at the J2 pads (net ties) (10.1.3, p.205).
5. Balancing heat: ~3 W/board at 4 channels; spread R_CB (2512), keep away from U1 and J3 (7.3.3.1.3, p.39).

Power



Cell Sense & Balance



Comments:	Company:		Variant:	
	Board Name: **BMS_SLAVE**		Project Name: **BMS Slave**	
Sheet Title: Cell Sense & Balance	File Name: cell_sense_and_balance.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
Sheet Path: /Project Architecture/Cell Sense & Balance/		Reviewer:	Size: **A3**	Sheet: **5** of **10**

[6] Power & Reference

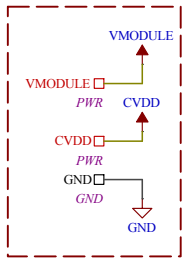
LAYOUT NOTE:
U1 GROUNDING (TI SLUSE81F §10.1.1)

All grounds are one net (GND) in the schematic. In layout, each bypass cap returns to its own ground pin through a small L1 copper island. Islands join the L2 GND plane through vias only. Do NOT connect islands to each other on L1.

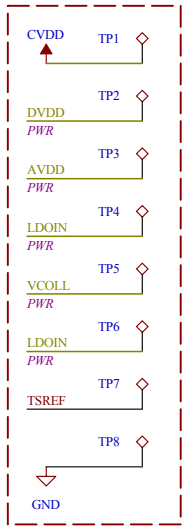
GROUND	PIN	CAPS RETURNING TO IT
AVSS	39	AVDD (38), BAT filter, VC0 filter, CB0 filter, thermal pad (65)
DVSS	50	DVDD (49)
CVSS	46	LDOIN (47), NEG5V (44), CVDD (45), TSREF (51), GPIO filter caps, daisy-chain bypass caps
REFHM	36	REFHP (37) ONLY - keep separate from AVSS on L1, join at L2 plane

LAYOUT NOTE:
RULES
1. Cap within 2 mm of its pin, same layer, no via between cap and pin.
2. Cap ground pad -> its island -> via to L2. Not a shared trace.
3. Keep pins 40-43 (COMH/COML) clear; place caps above and below the daisy-chain exit, not in it.
4. Thermal pad (65): 36 vias to L2 GND plane, tented/plugged.

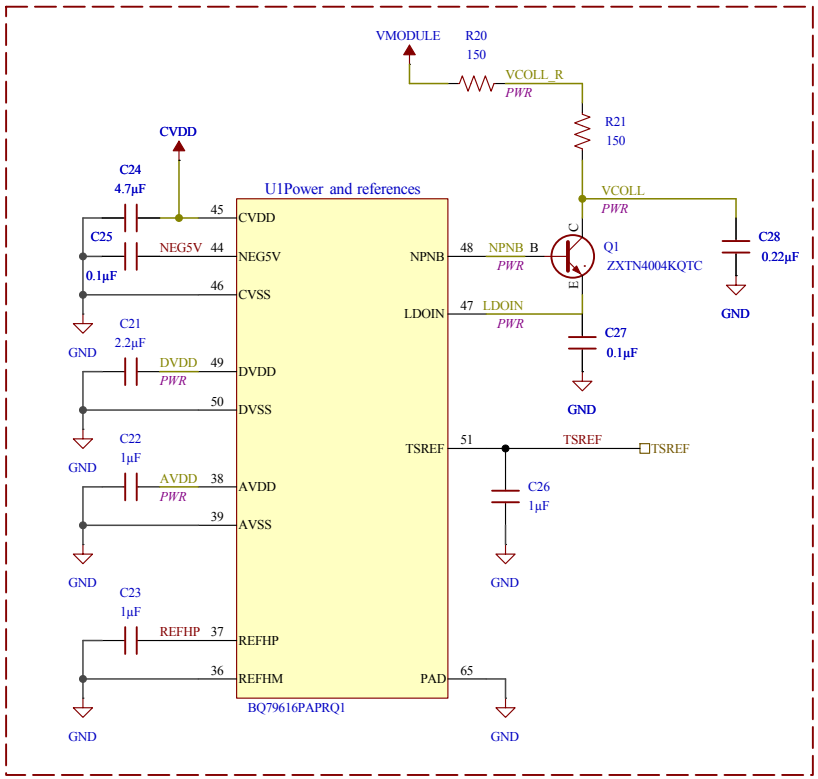
Power



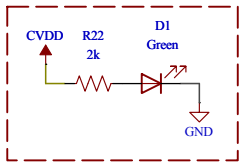
Test Points



Power and Reference



LED Indicator



LAYOUT NOTE: POWER (SLUSE81F)
1. Bypass caps as close as possible, REFHP most critical (10.1.2, p.205).
2. Ground groups per table; islands on L1, join on L2 (10.1.1, Fig. 10-1, p.203-204). NEG5V -> CVSS per 10.1.1 (Fig. 8-1 shows AVSS; same net).
3. Q1 DPAK: >= 4 cm2 VCOLL copper on L1+L4, stitched; away from U1 and R_CB. R20/R21 share dissipation (8.2.1.2.2, Fig. 8-3, p.193).
4. Thermal pad: PAP0064F land pattern, vias tented/plugged (pkg drawing notes 8-9, p.220).

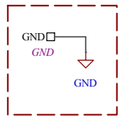
DESIGN NOTE:
CVDD LOAD BUDGET (SLUSE81F §7.3.1.5 p.23, Table 7-34)
Ext load max : 10 mA ACTIVE/SLEEP, 5 mA SHUTDOWN (ILIM 35 mA min)
RX tie : ~0 mA (input leakage)
LED + 2k0 : ~1.5 mA (only via JP1)
TOTAL : 1.5 mA of 10 mA -> OK
Daisy comm (~10 mA broadcast) is internal, not in ext budget.
NO other loads on CVDD. Temp sensors = TSREF, balancing = cells.
SHUTDOWN: CVDD not regulated, follows LDOIN (3.4-5.5 V) for WAKE.
IC draws 16-23 uA; LED would draw ~1 mA -> JP1 OPEN in car.

blue devil MOTORSPORTS

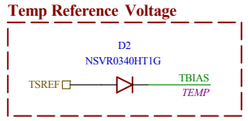
Comments:	Company:		Variant:	
	Board Name: **BMS_SLAVE**		Project Name: **BMS Slave**	
Sheet Title: Power & Reference	File Name: power_and_reference.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
Sheet Path: /Project Architecture/Power & Reference/		Reviewer:	Size: **A4**	Sheet: **6** of **10**

[7] Temperature Sense

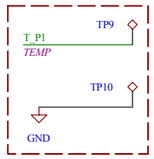
Power



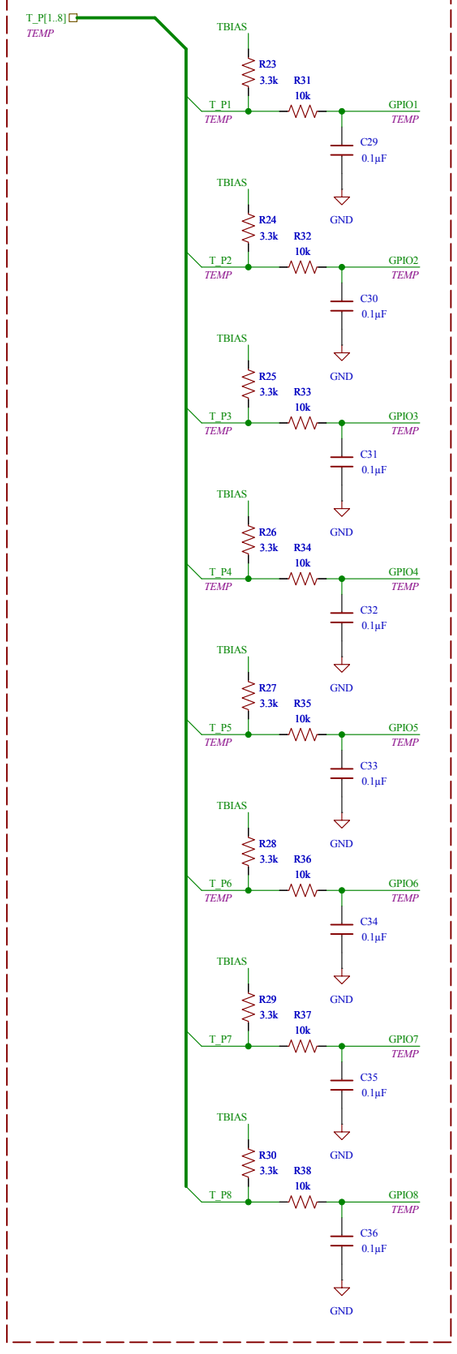
Temp Reference Voltage



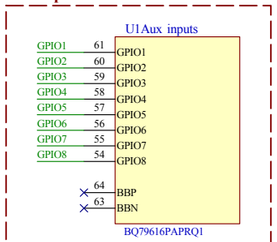
Test Points



Temp Reference Voltage



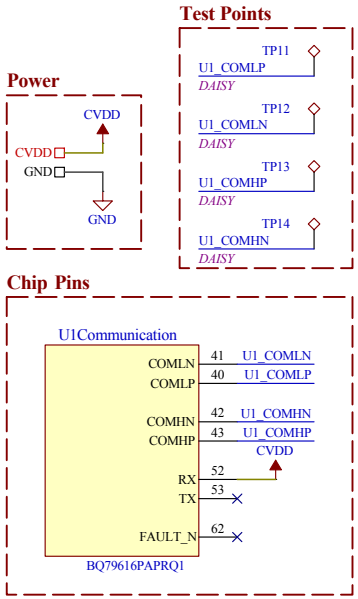
Aux Inputs

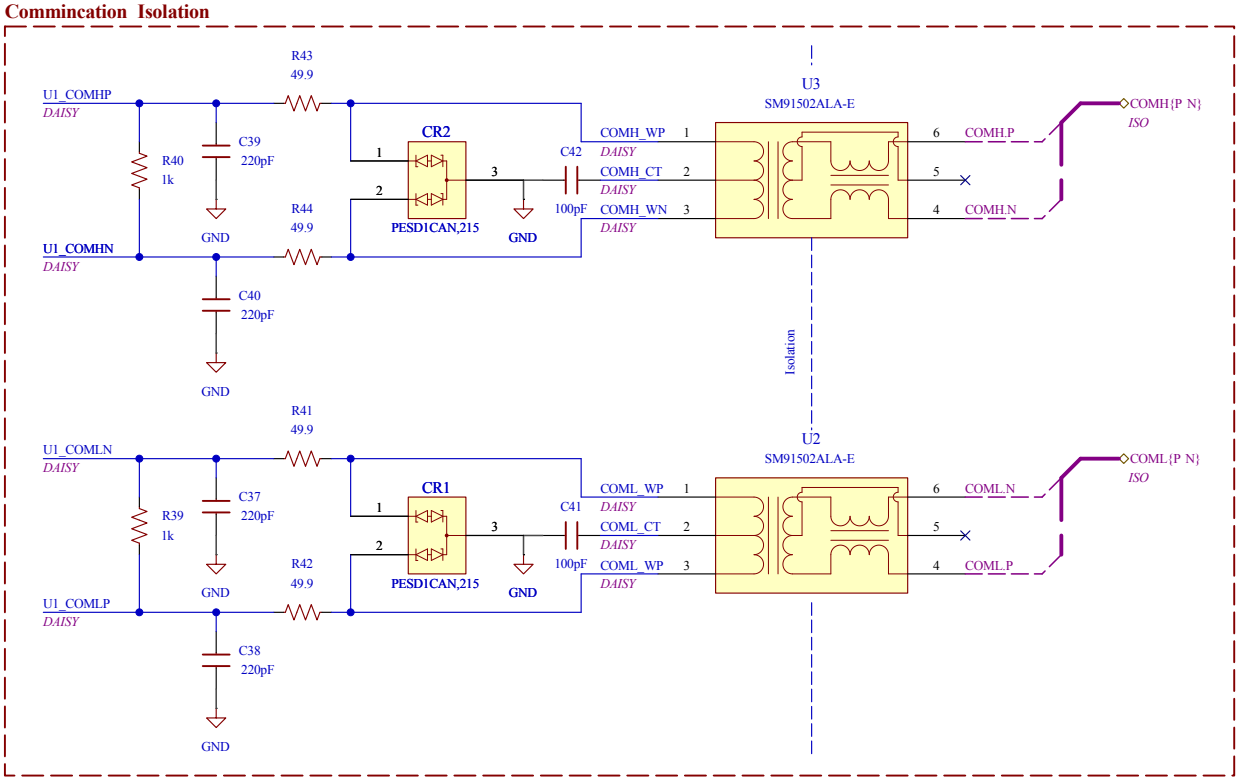


LAYOUT NOTE: POWER (SLUSE81F)
1. Bypass caps as close as possible, REFHP most critical (10.1.2, p.205).
2. Ground groups per table; islands on L1, join on L2 (10.1.1, Fig. 10-1, p.203-204). NEGSV -> CVSS per 10.1.1 (Fig. 8-1 shows AVSS; same net).
3. Q1 DPAK: >= 4 cm2 VCOLL copper on L1+L4, stitched; away from U1 and R_CB. R20/R21 share dissipation (8.2.1.2.2, Fig. 8-3, p.193).
4. Thermal pad: PAP0064F land pattern, vias tented/plugged (pkg drawing notes 8-9, p.220).

Comments:	Company:		Variant:	
	Board Name: **BMS_SLAVE**		Project Name: **BMS Slave**	
Sheet Title: Temperature Sense	File Name: temperature_sense.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
Sheet Path: /Project Architecture/Temperature Sense/		Reviewer:	Size: **A3**	Sheet: **7** of **10**

[8] Daisy Chain



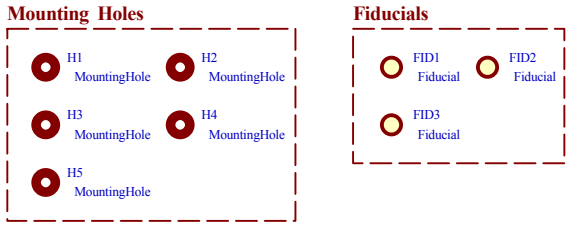


LAYOUT NOTE: DAISY CHAIN (SLUSE81F 10.1.4, Fig. 10-3, p.205-206)

1. P/N pairs short, straight, same layer, parallel, length matched.
2. R_TERM + 220 pF at U1 pins 40-43 -> CVSS. Values per Table 8-5 / Fig. 8-9 (p.198).
3. Transformer close to its connector; keep-out (no traces, no planes) around daisy-chain parts on ALL layers.
4. Barrier between transformer pin rows (1-3 | 4-6): 7.11 mm pad gap per Bourns SM91502ALA layout; no copper under body; cable side >= 3 mm to everything (FSAE EV.7.5.7 300-600 V row).
5. Same transformer PN on both ends of every link (TI SLVAEP4 5.1).

Comments:	Company:		Variant:	
	Board Name: **BMS_SLAVE**		Project Name: **BMS Slave**	
Sheet Title: Daisy Chain	File Name: daisy_chain.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
Sheet Path: /Project Architecture/Daisy Chain/		Reviewer:	Size: **A4**	Sheet: **8** of **10**

[9] Mechanical



Comments:	Company:		Variant:	
	Board Name: **BMS_SLAVE**		Project Name: **BMS Slave**	
Sheet Title: Mechanical	File Name: mechanical.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
Sheet Path: /Project Architecture/Mechanical/		Reviewer:	Size: **A4**	Sheet: **9** of **10**

[10] Revision History

Comments:	Company: **Blue Devil Motor Sports**		Variant:	
	Board Name: **BMS_SLAVE**		Project Name: **BMS Slave**	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Aidan Brzezinski	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Revision History/		Reviewer:	Size: **A4**	Sheet: **10** of **10**